

Development Guide

This section documents the technical structure, navigation architecture, HTML template, and wireframe for the Supercars website.

Site Directory Structure

The following directory structure organizes all assets for the Supercars site. Files are separated by type into dedicated subfolders for maintainability.

Path	Contents / Purpose
supercars/	Root directory contains all HTML files and subdirectories
images/	All JPG/PNG/WebP image files (car photos, logos, hero backgrounds)
styles/	CSS stylesheets styles.css (main), responsive.css (breakpoints)
fonts/	Any locally hosted font files (fallback for Bebas Neue, Inter)
trinity/	Holy Trinity deep-dive pages p1.html, laferrari.html, 918.html
brands/	Individual brand pages ferrari.html, lamborghini.html, mclaren.html, etc.
culture/	Car culture sub-pages (films, games, motorsport)
index.html	Home page hero, brands overview, Holy Trinity cards, sidebar
brands.html	Full brands hub page organized by national car culture
culture.html	Car culture in film, games, and motorsport
hobbies.html	Hot Wheels collecting, LEGO Technic builds, sim racing
trinity.html	Holy Trinity hub links to all three individual car deep-dives

Main Navigation

The site uses a sticky top navigation bar linking to all major sections. The navigation flow below shows each unique page and how they connect.

Page	Links To	Accessible From
index.html (Home)	brands.html, culture.html, hobbies.html, trinity.html	All pages via nav bar
brands.html	ferrari.html, lamborghini.html, mclaren.html, bugatti.html, koenigsegg.html, pagani.html	Home (nav), index brand list

trinity.html	p1.html, laferrari.html, 918.html	Home (nav + hero cards)
culture.html	External: IMDb, Wikipedia pages for films/games	Home (nav + sidebar)
hobbies.html	External: Hot Wheels official, LEGO Technic	Home (nav + sidebar)
trinity/*.html	Back to trinity.html; external: Google Image search for each car	trinity.html, Home hero cards
brands/*.html	Back to brands.html; external: manufacturer sites	brands.html

Navigation Flow (text outline):

- Home (index.html)
 - Brands (brands.html)
 - Ferrari, Lamborghini, McLaren, Bugatti, Koenigsegg, Pagani
 - Holy Trinity (trinity.html)
 - McLaren P1, Ferrari LaFerrari, Porsche 918 Spyder
 - Car Culture (culture.html)
 - Hobbies (hobbies.html)

HTML Template

The template below serves as the master structure for all pages on the site. Each section is self-documented with comments explaining its role. All pages are generated from this single template.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>Supercars [Page Title]</title>
  <link rel="stylesheet" href="../styles/styles.css" />
</head>
<body>

  <!-- IDENTITY BLOCK: logo + brand name, appears on every page -->
  <header class="site-header" role="banner">
    <div class="header-inner">
      <div class="site-identity">
        <span class="site-logo-icon" aria-hidden="true"> 🏎️ </span>
        <span class="site-logo-text">Supercars</span>
      </div>
      <!-- NAVIGATION: sticky, links to all main sections -->
      <nav aria-label="Main Navigation">
        <ul>
          <li><a href="../index.html">Home</a></li>
          <li><a href="../brands.html">Brands</a></li>
          <li><a href="../culture.html">Culture</a></li>
          <li><a href="../hobbies.html">Hobbies</a></li>
          <li><a href="../trinity.html">Holy Trinity</a></li>
        </ul>
      </nav>
    </div>
  </header>
```

```
<!-- MAIN CONTENT: primary page content goes here -->
<main id="main-content">
  <section aria-labelledby="page-heading">
    <h1 id="page-heading">[Page Heading]</h1>
    <p>[Page intro text]</p>
    <!-- Page-specific content blocks -->
  </section>
</main>

<!-- FOOTER: copyright, links, tagline -->
<footer class="site-footer" role="contentinfo">
  <div class="footer-inner">
    <span class="footer-logo">🏎️ Supercars</span>
    <nav class="footer-nav" aria-label="Footer Navigation">
      <a href="../index.html">Home</a>
      <a href="../brands.html">Brands</a>
      <a href="../trinity.html">Holy Trinity</a>
    </nav>
    <p class="footer-copy">&copy; 2025 Vaishvik Dogiparthi</p>
  </div>
</footer>

</body>
</html>
```

Wireframe

The wireframe consists of one plain HTML file per navigation section. Pages are unstyled (black and white, default browser fonts) but fully linked together. Each page states its section name and lists the content topics it will contain when complete.

File	Heading Text	Topics Listed on Page
index.html	Supercars Home	Hero banner, brand links, Holy Trinity cards, culture sidebar, hobbies
brands.html	Supercar Brands	Italian culture (Ferrari, Lamborghini, Pagani), British (McLaren), Swedish (Koenigsegg), French (Bugatti)
trinity.html	The Holy Trinity	McLaren P1, Ferrari LaFerrari, Porsche 918 Spyder specs, history, significance
culture.html	Car Culture	Films (Fast & Furious, Rush, Cars), TV (Formula 1), Games (NFS, Gran Turismo)
hobbies.html	Supercar Hobbies	Hot Wheels die-cast collecting, LEGO Technic car sets, sim racing

GitHub repository name: vaishvik-supercars | All wireframe pages link to each other via a plain <nav> block at the top.

Interactivity List (Bonus)

Interactive Element	Type	Purpose & Notes
Holy Trinity hover cards	CSS + JS event	On hover, dim the car image and reveal name + specs overlay. Keyboard-accessible via :focus-visible.
Favorite Car field	contenteditable	Visitors type their favorite car directly on the page no form submission needed. Simple entry point.
Hamburger nav menu	CSS checkbox trick	Mobile navigation toggle using a hidden checkbox input no JS required, accessible via label.
Brand spec comparison	JS table filter	Future: allow visitors to select two brands/cars and compare specs side-by-side in a dynamic table.
0–60 mph timer widget	JS countdown	Future: animated timer that counts down from 0 to the car's 0–60 mph time for each Trinity card.

SECTION 2

Design Guide

This section documents the visual design decisions for the Supercars site, including layout, color palette, typography, style rules, and media assets.

Layout & Storyboard

The site uses a two-column CSS grid layout on desktop (2/3 main content + 1/3 sidebar), collapsing to a single column on screens below 900px. The layout follows the rule of thirds: the hero occupies the top visual third, the content grid fills the middle, and the footer anchors the bottom.

[STICKY HEADER Logo | Brands Culture Hobbies Holy Trinity]

[HERO Full-width "Speed. Power. Beauty." tagline + H1]

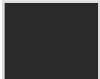
[CONTENT GRID | SIDEBAR]
[2/3 Brands, Holy Trinity cards, spec tables] | [1/3 Car Culture, Hobbies, Fav Car]

[FOOTER Logo + tagline | Nav links | Copyright]

Color Palette

The palette is built around a dark background with a racing red accent. Near-white body text ensures high contrast. Muted grays serve as secondary text and UI dividers.

Chip	Name	Hex	Usage
	Racing Red	#E10600	Primary accent nav underlines, headings, borders, logo text, footer bar
	Red Bright	#FF3322	Hover states brand card border, nav link hover, interactive highlights
	Midnight Black	#0A0A0A	Page background creates the high-contrast, high-performance atmosphere
	Carbon Dark	#141414	Card backgrounds Trinity cards, brand list items, interactive elements
	Asphalt	#111111	Header and footer background slightly lighter than page BG for depth
	Near White	#F0F0F0	Primary body text WCAG-level contrast against dark backgrounds
	Dim Gray	#AAAAAA	Secondary text subtitles, nav link default state, intro paragraphs
	Muted Gray	#888888	Footer text, metadata, captions, least-important UI labels

	Dark Border	#2A2A2A	Card borders, list dividers, UI separators subtle structure without brightness
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Style Guide

The following style rules define the visual presentation of the site's key elements. Full CSS is in styles/styles.css; this guide lists anticipated rules organized by component.

Typography

Element	Font / Size	Notes
Logo / Display headings	Bebas Neue, fallback Impact	All-caps, 1.7rem (header), 5.5rem (hero), 1.6rem (section h2)
Body text	Inter 400, 1rem / 1.7 line-height	Default text color #F0F0F0 on #0A0A0A background
Nav links	Inter 700, 0.85rem	Uppercase, letter-spacing 0.07em, color #AAA default → #FFF hover → #E10600 active
Section headings (h2)	Bebas Neue, 1.6rem	Color #E10600, border-bottom 1px rgba(225,6,0,0.3), uppercase
Card overlay text	Bebas Neue 1.1rem / Inter 0.75rem	Car name in Bebas Neue white; stats in Inter semi-transparent white
Footer / metadata	Inter 400, 0.75–0.8rem	Color #888, uppercase, letter-spacing 0.1em

Layout & Spacing

- Max content width: 1100px, centered with auto margins
- Gutter / padding: 1.5rem on mobile; scales up with clamp()
- Content grid: grid-template-columns: 2fr 1fr (desktop) → 1fr (≤900px)
- Sidebar: position: sticky, top: 60px + 1.5rem (below header)
- Hero padding: clamp(3rem, 10vw, 6rem) top and bottom

Header & Navigation

- Header: position: sticky, top: 0, z-index: 100, background #111, border-bottom 3px solid #E10600
- Nav links: underline animation via ::after pseudo-element, left/right transition from center
- Active link: color #E10600, underline visible at rest
- Mobile (≤768px): hamburger toggle appears, nav collapses to vertical list

Cards (Holy Trinity)

- Aspect ratio: 4/3, overflow: hidden, border-radius: 6px
- Hover: image scale(1.06) + brightness(0.4), overlay opacity 0 → 1
- Focus: box-shadow 0 0 0 3px #E10600 for keyboard accessibility

Sidebar Sections

- Background: #141414, border: 1px solid #2A2A2A, border-left: 3px solid #E10600
- Sticky below header, drops to static on mobile (≤900px)

- Culture list items separated by 1px border-bottom #2A2A2A
- contenteditable field: dashed red border, turns solid on :focus

Media Assets

The following image and media files are used or planned for the Supercars site. Lazy loading is applied to all below-fold images.

File / Description	Type	Source	License / Notes
McLaren P1 front 3/4 view	JPG	sportscarmarket.com	Editorial use; to be replaced with CC-licensed photo before publish
Ferrari LaFerrari in red	JPG	cdn.shopify.com (dealer)	Editorial placeholder; replace with Wikimedia Commons CC image
Porsche 918 Spyder on road	JPG	motortrend.com	Editorial placeholder; Porsche press images available via press.porsche.com
Hero background texture	CSS gradient	Generated (no external file)	Pure CSS radial + linear gradient no licensing concern
Brand logo images (future)	SVG/PNG	Wikipedia / official press kits	Use only official press-kit logos; verify trademark use policy per brand
Car culture posters (future)	JPG/PNG	Movie studio press kits	Fast & Furious / Rush press images; verify editorial-use terms
Bebas Neue font	WOFF2	Google Fonts	Open Font License (OFL) free for all use
Inter font	WOFF2	Google Fonts / rms.me	Open Font License (OFL) free for all use

APPENDIX

Appendix Project Overview

Field	Detail
Project Name	Supercars
Developer	Vaishvik Dogiparthi
Site Tagline	Speed. Power. Beauty.
Primary Goal	Educate and inspire automotive enthusiasts with a visually compelling, personal take on the supercar world
Target Audience	Supercar enthusiasts (15–40), casual car fans, pop-culture viewers, collectors and hobbyists
Site Personality	Bold, high-performance, dark aesthetic energized and exciting without being unwelcoming
Tech Stack	HTML5, CSS3 (custom properties, CSS Grid, Flexbox), vanilla JS for interactions
Responsive Breakpoints	≤900px: single column layout; ≤768px: hamburger nav; ≤540px: trinity cards stack
Comparable Sites	Motor Trend (motortrend.com), Top Gear (topgear.com), Supercar Blondie (supercarblondie.com)
GitHub Repo	vaishvik-supercars